SDLC AND USE CASE DIAGRAM DESIGN FOR EDUVERSE

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1. SDLC Model Matrix

Criteria	Priority	Waterfall	V-Model	Iterative	Incremental	Spiral	Agile
Requirement flexibility	5	2	2	4	4	4	5
User feedback	5	1	2	4	4	4	5
Early delivery	4	1	2	4	5	4	5
Risk management	3	2	4	3	3	5	4
Continuous improvement	4	1	2	5	4	4	5

Selection: Agile is the most suitable model for EduVerse because it provides flexibility, continuous feedback, incremental delivery, and continuous improvement.

2. Justification:

For the EduVerse project, we selected the Agile Model based on the comparison table. The main reasons are:

Requirement Flexibility: EduVerse may require new features or changes during development. Agile allows requirements to be changed easily in different iterations.

User Feedback: Students, instructors, and administrators are the main users of EduVerse. Agile allows their feedback to be collected continuously and incorporated into the system.

Early Delivery: The platform can be developed and delivered in small parts. For example, registration and login can be developed first, followed by course enrollment, video lectures, quizzes, and assignments.

Continuous Improvement: After each iteration, the system can be reviewed and improved. Bugs and usability problems can be fixed before the next iteration.

Risk Management: Agile identifies problems early through frequent testing and review, reducing the risk of major problems at the end of the project.

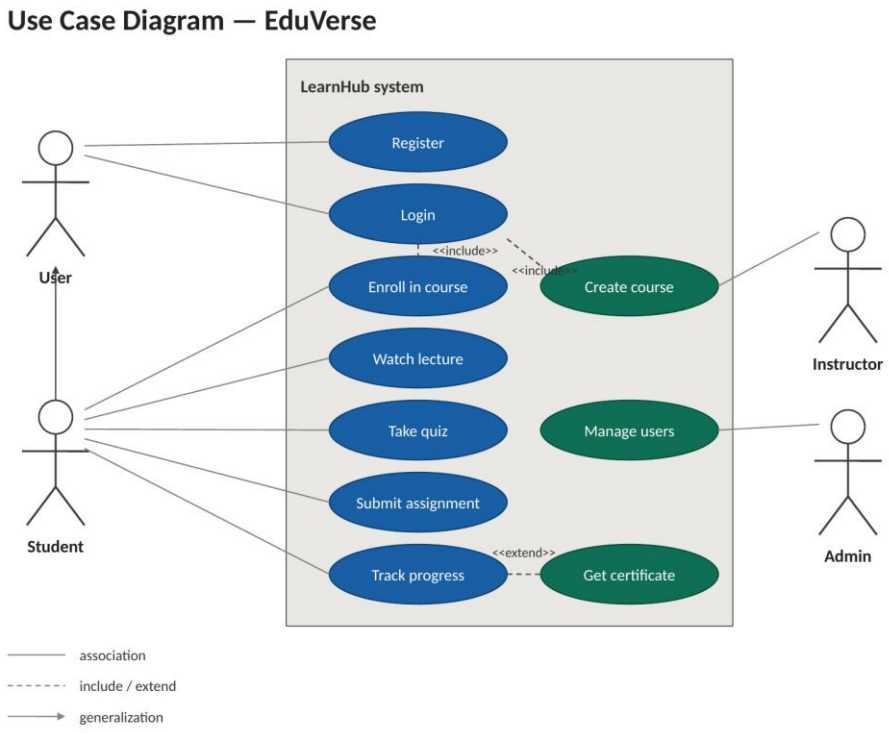


Figure 2: Use Case Diagram of EduVerse.

3. Description

Two relationship types beyond simple association are shown. The include relationship applies where a use case cannot execute without another: both Enroll in course and Create course include Login, since a user must be authenticated before performing either action. The extend relationship applies where a use case optionally builds on another under a condition: Get certificate extends Track progress, since a certificate is only generated once a student's tracked progress reaches completion, rather than as a mandatory outcome of every progress check.

Together, the diagram illustrates Eduverse core functional scope like authentication, course enrollment and consumption, assessment, progress monitoring, certification, course authoring, and platform administration and how responsibility for each function is distributed across its user roles.

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4.Conclusion

EduVerse provides a complete online learning environment. Agile is suitable because it supports flexible requirements, continuous feedback, and incremental development. The Use Case Diagram shows the main actors and their interactions with the system.